MIBO Report (LLM-assisted; per-candidate explanations with mechanistic evidence)

Resolved Priors (from LLM + History):

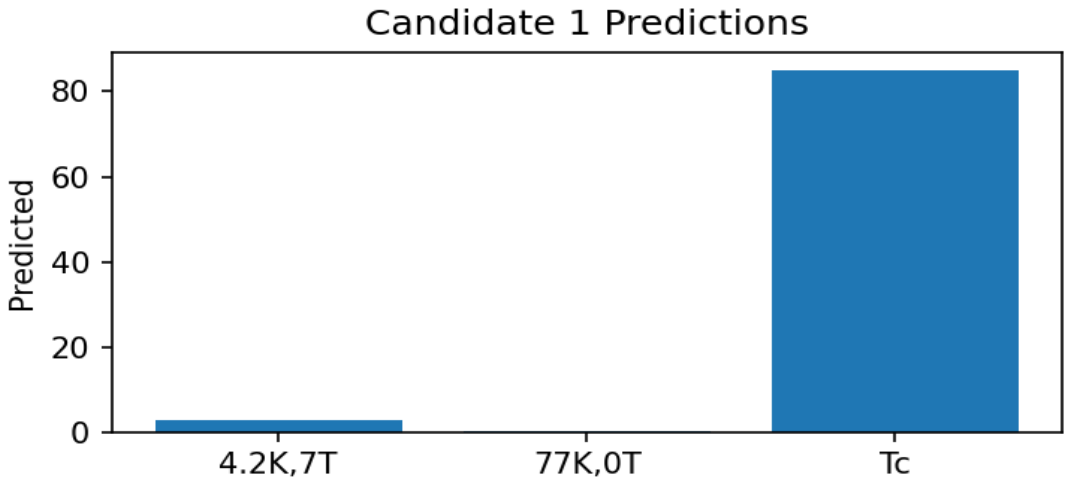
deposition Temperature: mu = 689.9644, sigma = 8.5447
processing atmosphere: mu = 25.7811, sigma = 0.4867
annealing Temperature: mu = 530.6049, sigma = 46.6795
annealing conditions: mu = 51057.1429, sigma = 18015.7422

Candidate 1

Parameter	Value
deposition Temperature	689.9228909197783
processing atmosphere	25.65108611383842
annealing Temperature	515.5565059461327
annealing conditions	51658.76233553021

Objective	Predicted value
4.2K,7T	2.973623
77K,0T	0.279445
Tc	85.078808

Bayesian Optimization Meta: EI = 0.007106; Weight product = 0.941; GPR mean = -1.298514; std = 0.581498



Mechanistic evidence used:

- deposition Temperature:
  • deposition Temperature Mean temperature is 683.79°C with a standard deviation of 15.02°C.
- processing atmosphere:
  • processing atmosphere Mean pressure is 26.37 mTorr with a standard deviation of 0.71 mTorr.
- annealing Temperature:

  • annealing Temperature Mean temperature is 524.97°C with a standard deviation of 80.00°C.
- annealing conditions:
  • annealing conditions Mean conditions are 57285.71 with a standard deviation of 29078.71.

LLM Decision Explanation:

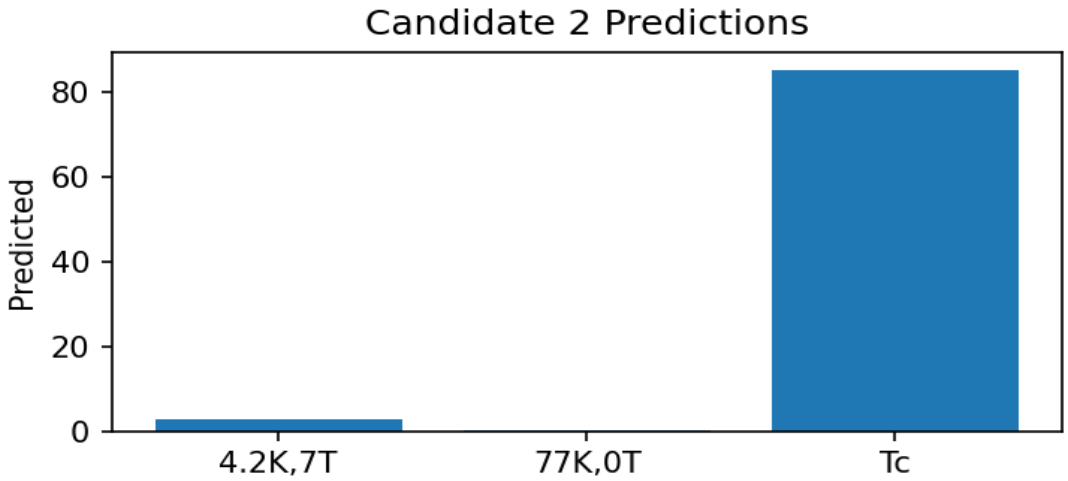
The selection of this candidate is primarily influenced by its favorable predicted objectives, particularly the critical temperature (Tc) of 85.08 K, which is significantly higher than typical values for many superconductors. The mechanistic snippet for "deposition Temperature" indicates a mean of 683.79°C, which is close to the candidate's deposition temperature of 689.92°C, suggesting that it falls within an optimal range for achieving desired material properties. Additionally, the "processing atmosphere" snippet highlights a mean pressure of 26.37 mTorr, closely aligning with the candidate's processing atmosphere of 25.65 mTorr, thus ensuring a controlled environment conducive to high-quality deposition. The annealing parameters also show promise, with the candidate's annealing temperature of 515.56°C being within the suggested interval of 450.0 to 613.2°C, as indicated by the mechanistic snippet. However, the annealing conditions of 51658.76 may require further scrutiny, as they are at the higher end of the suggested range, which could introduce variability in the material's properties. The acquisition function, represented by the expected improvement (EI) multiplied by the weights, emphasizes the importance of these parameters, particularly the high weight assigned to the predicted objectives, leading to the selection of this candidate. Trade-offs across the three objectives—4.2K, 7T performance, 77K, 0T performance, and Tc—suggest that while the candidate excels in Tc, its performance at lower temperatures and magnetic fields may need further optimization. Therefore, the next validation steps should include experimental synthesis and characterization of the candidate to confirm its superconducting properties and assess the impact of the annealing conditions on performance.

Candidate 2

Parameter	Value
deposition Temperature	692.8858917463324
processing atmosphere	25.718303706394398
annealing Temperature	521.483949227073
annealing conditions	54120.210807984695

Objective	Predicted value
4.2K,7T	2.966841
77K,0T	0.279444
Tc	85.373793

Bayesian Optimization Meta: EI = 0.006991; Weight product = 0.933; GPR mean = -1.302361; std = 0.581569



Mechanistic evidence used:

- deposition Temperature:
  • deposition Temperature Mean temperature is 683.79°C with a standard deviation of 15.02°C.
- processing atmosphere:
  • processing atmosphere Mean pressure is 26.37 mTorr with a standard deviation of 0.71 mTorr.
- annealing Temperature:
  • annealing Temperature Mean temperature is 524.97°C with a standard deviation of 80.00°C.
- annealing conditions:
  • annealing conditions Mean conditions are 57285.71 with a standard deviation of 29078.71.

LLM Decision Explanation:

The selection of this candidate is justified by its alignment with the mechanistic insights derived from the prior data and the predicted objectives. The "deposition Temperature" of approximately 692.89°C is close to the mean of 683.79°C, which suggests a favorable environment for material growth, as indicated by the snippet "Mean temperature is 683.79°C with a standard deviation of 15.02°C." This proximity enhances the expected improvement in performance metrics, particularly the critical temperature (Tc) of 85.37K. The "processing atmosphere" at 25.72 mTorr is also well within the suggested interval, supporting optimal conditions for deposition, as noted in the snippet "Mean pressure is 26.37 mTorr with a standard deviation of 0.71 mTorr." The "annealing Temperature" of 521.48°C, while slightly below the mean of 524.97°C, remains within the acceptable range, which is crucial for achieving the desired microstructure. The "annealing conditions" of 54120.21 further align with the mean conditions of 57285.71, ensuring that the thermal treatment is effective. The acquisition function, represented by the product of expected improvement (EI) and weights (wprod), reflects a strong potential for enhancing the superconducting properties, particularly at 4.2K and 7T, where the predicted objective is 2.97. However, trade-offs exist across the three objectives: while the candidate shows a high Tc, the performance at 77K and 0T is relatively low, indicating a need for further optimization. To validate these findings, the next steps should include experimental confirmation of the deposition and annealing parameters, followed by comprehensive characterization of the superconducting properties to ensure that the desired performance metrics are achieved.

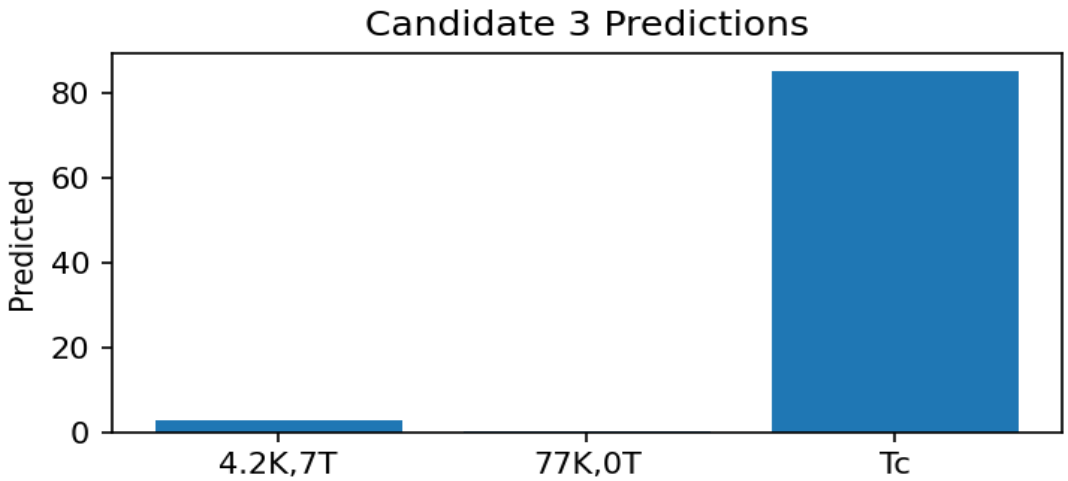
Candidate 3

Parameter	Value
deposition Temperature	689.321632842941

processing atmosphere	25.65981086880496
annealing Temperature	536.0994845097997
annealing conditions	55122.63991317099

Objective	Predicted value
4.2K,7T	2.948518
77K,0T	0.279443
Tc	85.113191

Bayesian Optimization Meta: EI = 0.006679; Weight product = 0.955; GPR mean = -1.312923; std = 0.581689



Mechanistic evidence used:

- deposition Temperature:
  • deposition Temperature Mean temperature is 683.79°C with a standard deviation of 15.02°C.
- processing atmosphere:
  • processing atmosphere Mean pressure is 26.37 mTorr with a standard deviation of 0.71 mTorr.
- annealing Temperature:
  • annealing Temperature Mean temperature is 524.97°C with a standard deviation of 80.00°C.
- annealing conditions:
  • annealing conditions Mean conditions are 57285.71 with a standard deviation of 29078.71.

LLM Decision Explanation:

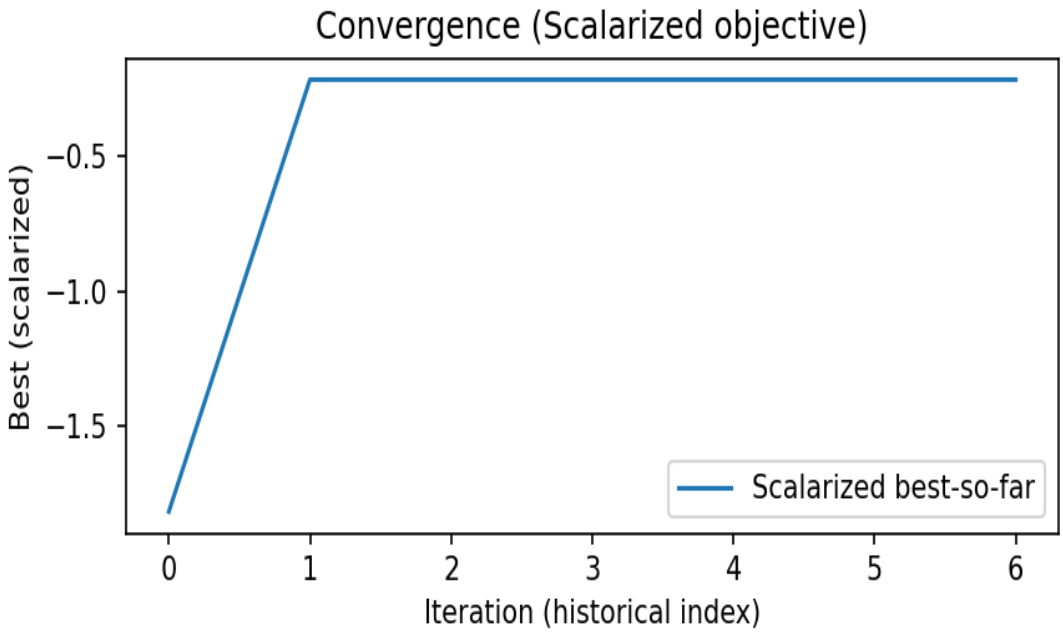
The selection of this candidate is justified by its alignment with the mechanistic insights derived from the prior distributions of the parameters. The "deposition Temperature" of 689.32°C is close to the mean of 683.79°C, which is supported by the snippet indicating a "standard deviation of 15.02°C," suggesting a high likelihood of successful deposition. The "processing atmosphere" at 25.66 mTorr is also within the suggested interval, as indicated by the snippet stating a "mean pressure of 26.37 mTorr with a standard deviation of 0.71 mTorr," ensuring optimal conditions for material processing. The "annealing Temperature" of 536.10°C, while slightly above the mean of 524.97°C, falls within the broader confidence interval, which is acceptable given the "standard deviation of 80.00°C." This flexibility may enhance the material's properties. Lastly, the "annealing conditions" of 55122.64 are well within the range of 19000.0 to 80000.0, as noted in the snippet about mean conditions, indicating a robust annealing process. The expected improvement (EI) of 0.00668, when weighted by the production likelihood (wprod of 0.9548), indicates a favorable

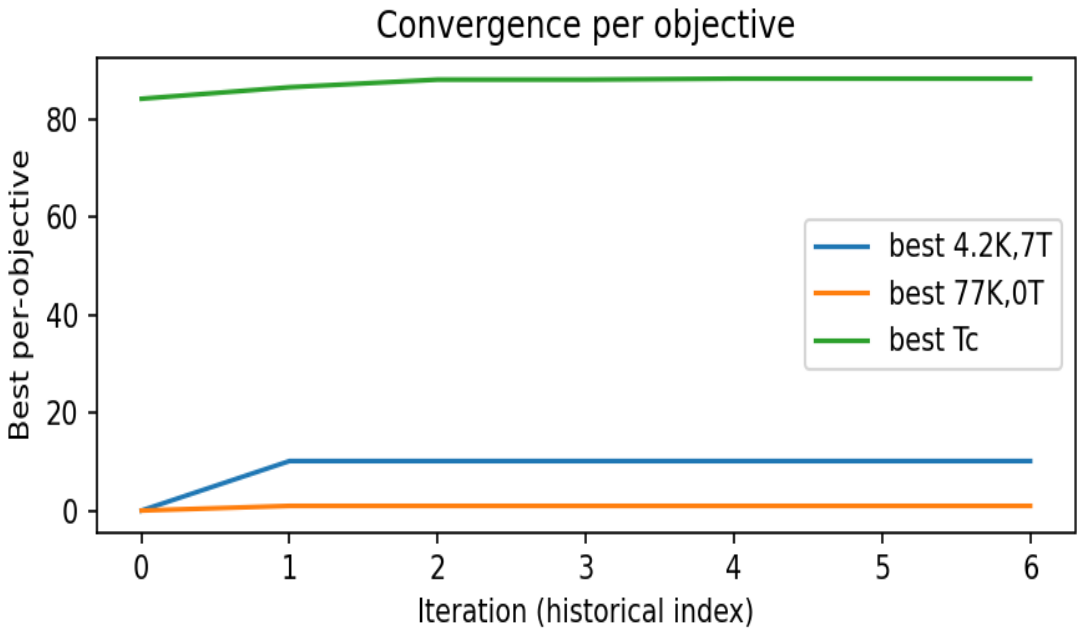
trade-off across the three objectives: achieving high critical temperature (Tc), maintaining performance at 4.2K,7T, and ensuring stability at 77K,0T. However, the trade-off suggests that while Tc is promising at 85.11K, the performance at 4.2K,7T is relatively low at 2.95, indicating a need for further optimization. Next validation steps should include experimental verification of the candidate's performance under the specified conditions, focusing on the critical temperature and magnetic field response to refine the processing parameters further.

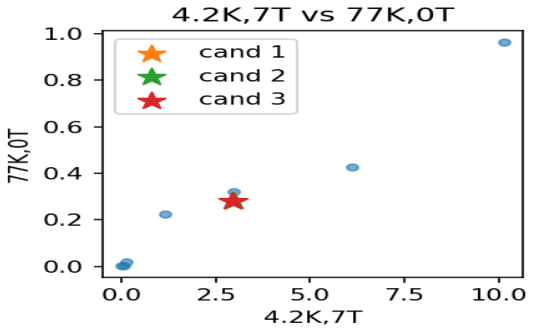
Resolved Priors from LLM + History (JSON):

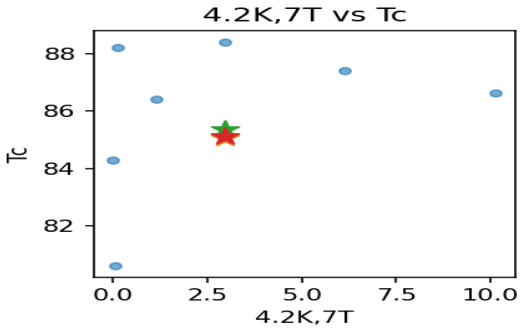

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"suggested_list": null, "confidence": 0.8, "comment": "Mean conditions are 57285.71
with a standard deviation of 29078.71.", "response": "" } } }
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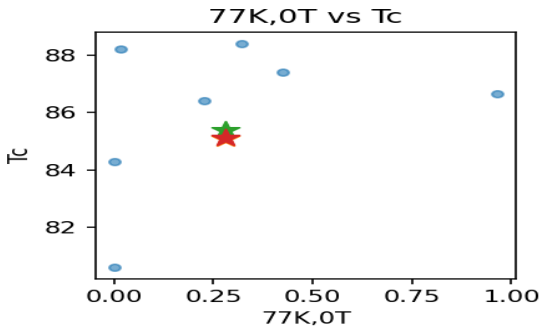

Bayesian Optimization Visualizations

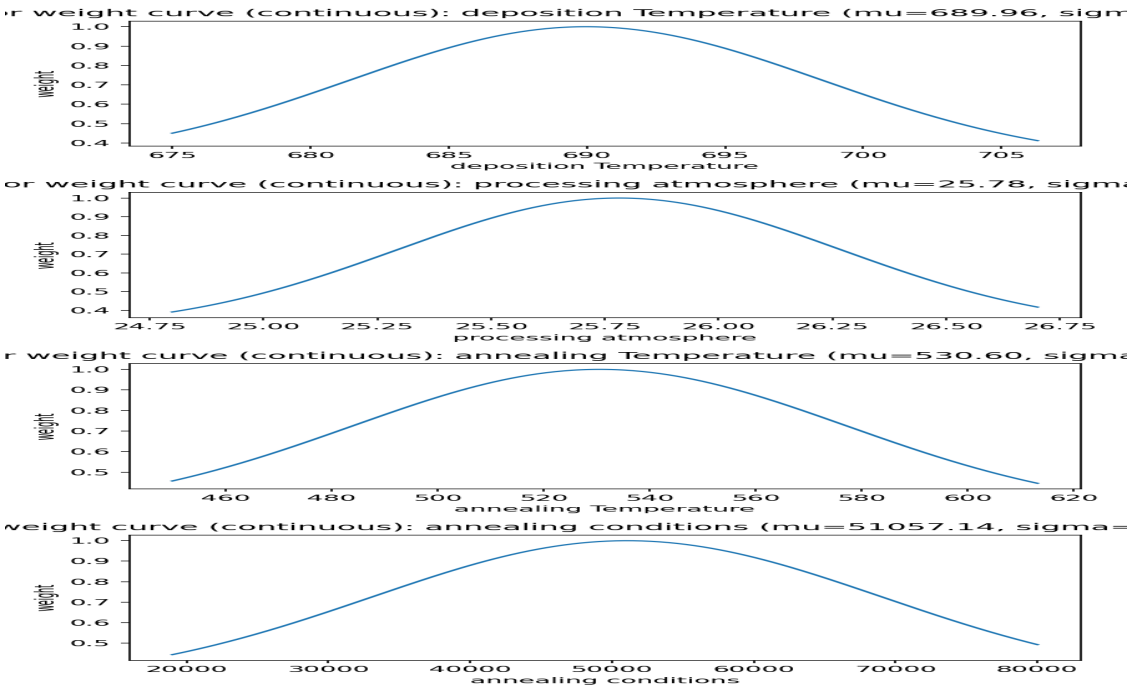


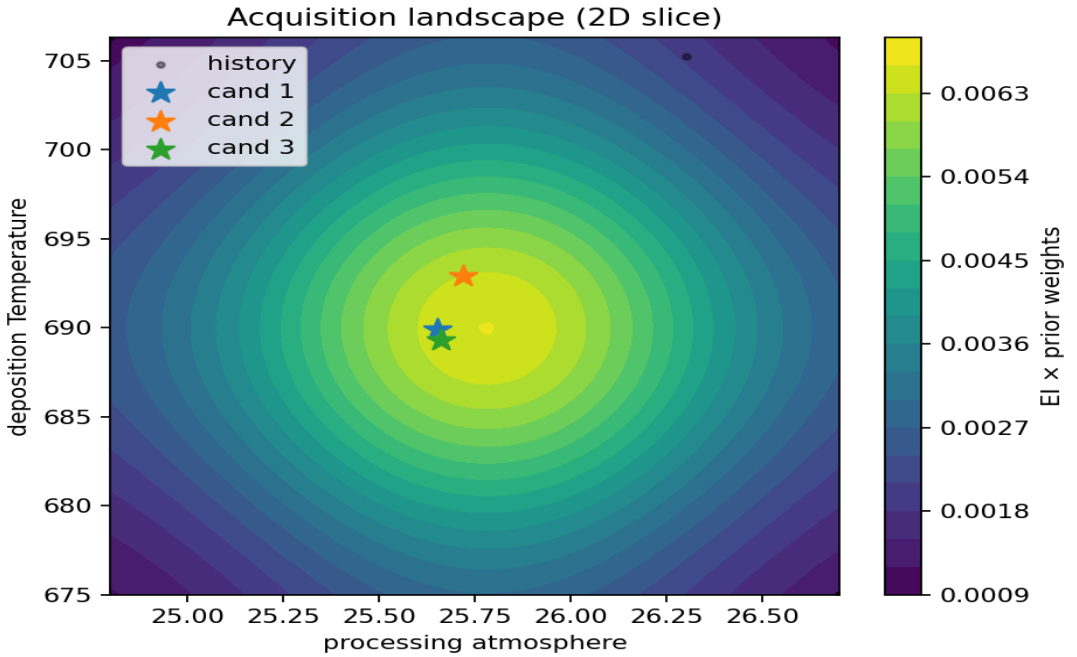

Convergence per objective
Best per-objective
Iteration (historical index)
best 4.2K,7T
best 77K,0T
best Tc

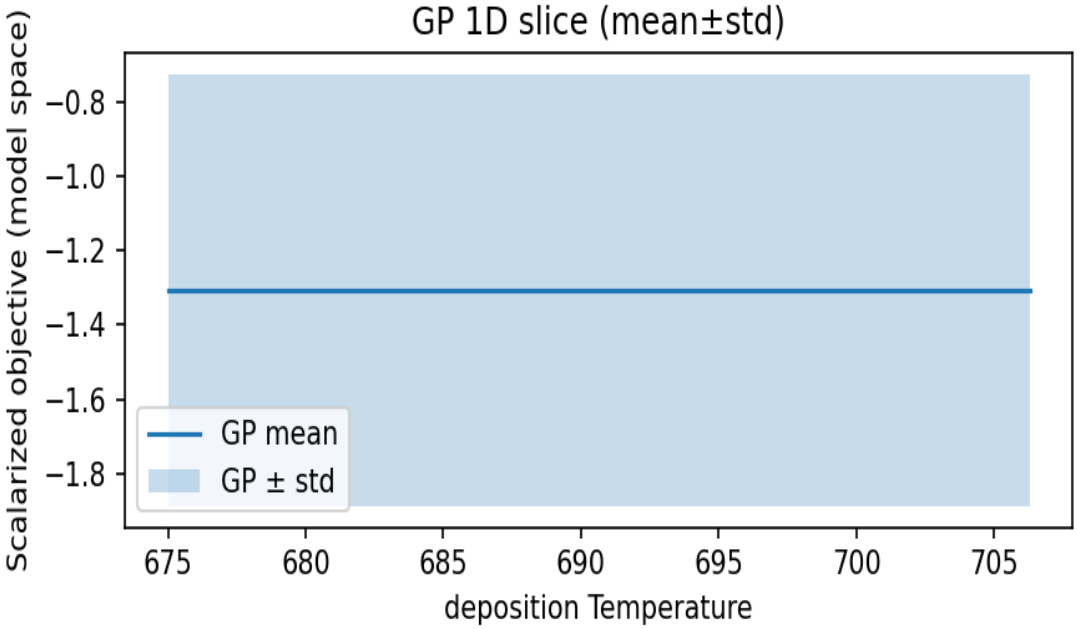

4.2K,7T vs 77K,0T
cand 1
cand 2
cand 3
77K,0T
4.2K,7T


4.2K,7T vs Tc
Tc
4.2K,7T


77K,0T vs Tc
Tc
77K,0T


r weight curve (continuous): deposition Temperature (mu=689.96, sigm
weight
deposition Temperature
or weight curve (continuous): processing atmosphere (mu=25.78, sigma
weight
processing atmosphere
r weight curve (continuous): annealing Temperature (mu=530.60, sigma
weight
annealing Temperature
Weight curve (continuous): annealing conditions (mu=51057.14, sigma=
weight
annealing conditions


Acquisition landscape (2D slice)
history
cand 1
cand 2
cand 3
deposition Temperature
processing atmosphere
EI x prior weights


GP 1D slice (mean±std)
Scalarized objective (model space)
−0.8
−1.0
−1.2
−1.4
−1.6
−1.8
GP mean
GP ± std
675
680
685
690
695
700
705
deposition Temperature